Forecasting for the Grain the Decision Is Made At: A Retail World Model on M5, Dominick's and Breakfast at the Frat

Karthik Iyer

Correspondence: karthik.iyer488@gmail.com ORCID: 0009-0004-0593-1602

Abstract. Which forecasting model is most accurate for a retailer depends on the level of the hierarchy the forecast is judged at, and the answer is different again on data where promotions drive sales. We compare a world model, which carries each item's state forward one period at a time under the price and promotion plan, with LightGBM, a lift model, a graph model, seasonal naive, and the pretrained TimesFM 2.5, which reads only past sales. Every model is scored at every level of the hierarchy on M5 (four months), Dominick's (six categories) and Breakfast at the Frat, with settings locked beforehand. *On M5, item by store,* the world model is more accurate than LightGBM in all four months (WRMSSE 0.851 against 0.873 in March, 0.897 against 0.905 in the official test month), and TimesFM is slightly more accurate than both (0.005 to 0.014 below the world model). *On M5, at the totals,* the world model has the lowest error in two of the four months, LightGBM in one, the official test month, and the graph model in one; a comparison on the test month alone would have ranked LightGBM first. TimesFM forecasts 5% to 10% below total sales, and from store by department up to the chain total the world model's error is 21% to 47% lower than TimesFM's. *On Dominick's and Frat*, where promoted weeks carry 18% to 61% of units, the pattern reverses: LightGBM is more accurate than the world model item by store in six of the seven data sets and level in the seventh, the world model is more accurate over all levels in two of six categories (soft drinks, analgesics) and LightGBM in three and on Frat, and TimesFM, which cannot see a promotion coming, forecasts 29% to 63% of promoted-week sales. *For an item's first price cut in a year,* which sold 120 units, the world model forecast 121, 600, 243 across seeds, LightGBM about 3 and TimesFM 8. A retailer should therefore choose the model by the decision it feeds, at that decision's level, compared over several periods. Every model and change that lost is reported.

Keywords: retail forecasting; hierarchical evaluation; promotions; world models; foundation models; M5.

1 Introduction

A retailer uses one sales forecast for several decisions, and those decisions are made at different levels of the product and store hierarchy. Price changes, promotions, supplier funding and store orders are decided for one item in one store, week by week. Sales, unit and margin plans are set and reviewed for a category or a whole chain, by month. The financial plan is a company total. A forecast can be accurate item by store and still miss the month's category total by several percent, and the reverse: small errors in the same direction on many items add up at the top of the hierarchy, while large errors in opposite directions cancel.

Forecasting competitions and most published comparisons rank models on one combined score over one test period. The M5 competition [1] already weights its score across twelve levels of the Walmart hierarchy, but reports one number per entry, for one 28-day month. This paper asks three questions that a single number hides: which model is most accurate at each level, whether that answer holds from one month to the next, and how much it depends on whether the model sees the plan (the prices, promotions, displays and events set in advance for the weeks being forecast).

We answer them with seven models on three public data sets. The models range from seasonal naive to a pretrained time-series foundation model (TimesFM 2.5 [36]) used without training on our data, and include a world model of our own: each item's recent history is turned into a state, and that state is carried forward one period at a time under the plan, so that what is planned for one week can change the forecast for the weeks after it (Figure 2). Figure 1 sets out the whole study. Before any final test we ran a development study on windows that precede the final weeks; every configuration in it is reported here, with the ones that lost.

The paper finds and contributes the following.

1. **The most accurate model depends on the level.** On M5, TimesFM given only sales history is the most accurate item by store in all four months tested, by 0.005 to 0.014 against the world model and 0.018 to 0.037 against LightGBM, while models that see the

plan are more accurate at every level from store by department up to the chain total (§6.1).

2. **One test month is not enough.** The official M5 test month is the only one of four in which LightGBM is the most accurate over all levels. The world model is lowest in March and, by 0.002, in April, and the graph model on the practice window. Between the world model and LightGBM, the month's winner is whichever gets the month's total closer (§6.2).
3. **The value of the plan depends on how much of the sales it moves.** Where prices rarely change (M5), giving TimesFM the plan changes its item-by-store error by at most 0.002. Where promoted weeks carry 18% to 61% of a category's units (Dominick's), TimesFM without the plan forecasts 29% to 63% of promoted-week sales, and with it 60% to 113% (§6.3).
4. **Plan-aware models respond to price changes that history has not shown, with a size the data do not fix.** For an item's first price cut in a year the world model forecast 121, 600, 243 units across seeds against 120 sold, where models working from history forecast under 10 (§6.4).
5. **A merchant-readable development record.** The development study finds, among other things, that most of a snack-cracker promotion's lift is taken from other crackers while analgesic and frozen-entree promotions mostly add to the category (§A.2); that a pretrained latent state carries most of the forecast signal without fine-tuning (§A.1); and that development-window rankings carried to the final weeks in some places and reversed in others (§6.5).
6. **Every model and change that lost is reported**, with its cause where one was measured (§7, Table 6).

The paper is organised as follows. Section 2 places the work among retail promotion models, hierarchical evaluation, global and pretrained forecasters, and world models. Section 3 describes the data and the levels scored. Section 4 gives the models and Section 5 the protocol. Section A reports the development study, Section 6 the final tests, and Section 7 the failures. Sections 8 to 10 discuss what the results mean for a retailer, their limits, and conclude.

2 Related work

2.1 Retail forecasting and the promotion plan

Fildes, Ma and Kolassa [5] review retail forecasting research and practice and note that most studies forecast and evaluate at one level of aggregation, usually the item, although retail decisions are spread across levels. Sales of a single item in a single store are intermittent: most items sell nothing on most days, which led to dedicated methods for intermittent demand [6, 7] and to evaluating forecasts as count distributions [8]. The Tweedie family [9], a compound Poisson–gamma distribution with a point mass at zero, is the loss many M5 entries used for this reason, and the loss we use.

Promotions move retail sales more than anything else in the plan. PromoCast forecast promoted-week sales for a chain from the planned event and the item's past response [10]. Van Heerde, Leeflang and Wittink [11] and Macé and Neslin [12] measured the dips in sales before and after a promotion from store scanner data, when shoppers stock up and then buy less. Ma, Fildes and Huang [13] showed that promotional information from within and across categories improves item-level forecasts. Where the extra sales of a promoted item come from is a long-standing question: Gupta [14] split the promotion bump into brand switching, purchase timing and quantity, and van Heerde, Gupta and Wittink [15] showed that measured in units, only about a third of the bump is switching from other brands. Attraction (market-share) models make an item's sales the category's sales times its share, with the share moved by price and promotion [16, 17]; our brand-switching step (§4) follows that structure. Dominick's Finer Foods ran pricing experiments across its stores [18], and its scanner data are behind much of this work [19].

2.2 Hierarchies and evaluation

Forecasts that must add up across a hierarchy are studied as hierarchical forecasting, from bottom-up and top-down to optimal reconciliation [21, 22, 23]. We do not reconcile: each model forecasts item by store and its forecasts are summed, which is how the M5 accuracy competition scored entries, with a weighted, scaled error averaged over twelve levels [1]. The scaling by each series' in-sample error follows Hyndman and Koehler's mean absolute scaled error [24]. The M5 data are a reasonable sample of retail series [3], and a companion competition scored forecast distributions [2]. Tashman [25] sets out why a comparison should use several forecast origins rather than one test period, which §6.2 confirms on M5.

2.3 Global models, trees and neural forecasters

Models fitted across many series at once, global models, are often more accurate than one model per series [26]. In M5 the leading entries were built on gradient-boosted trees [27, 28]; the winner averaged direct and recur-

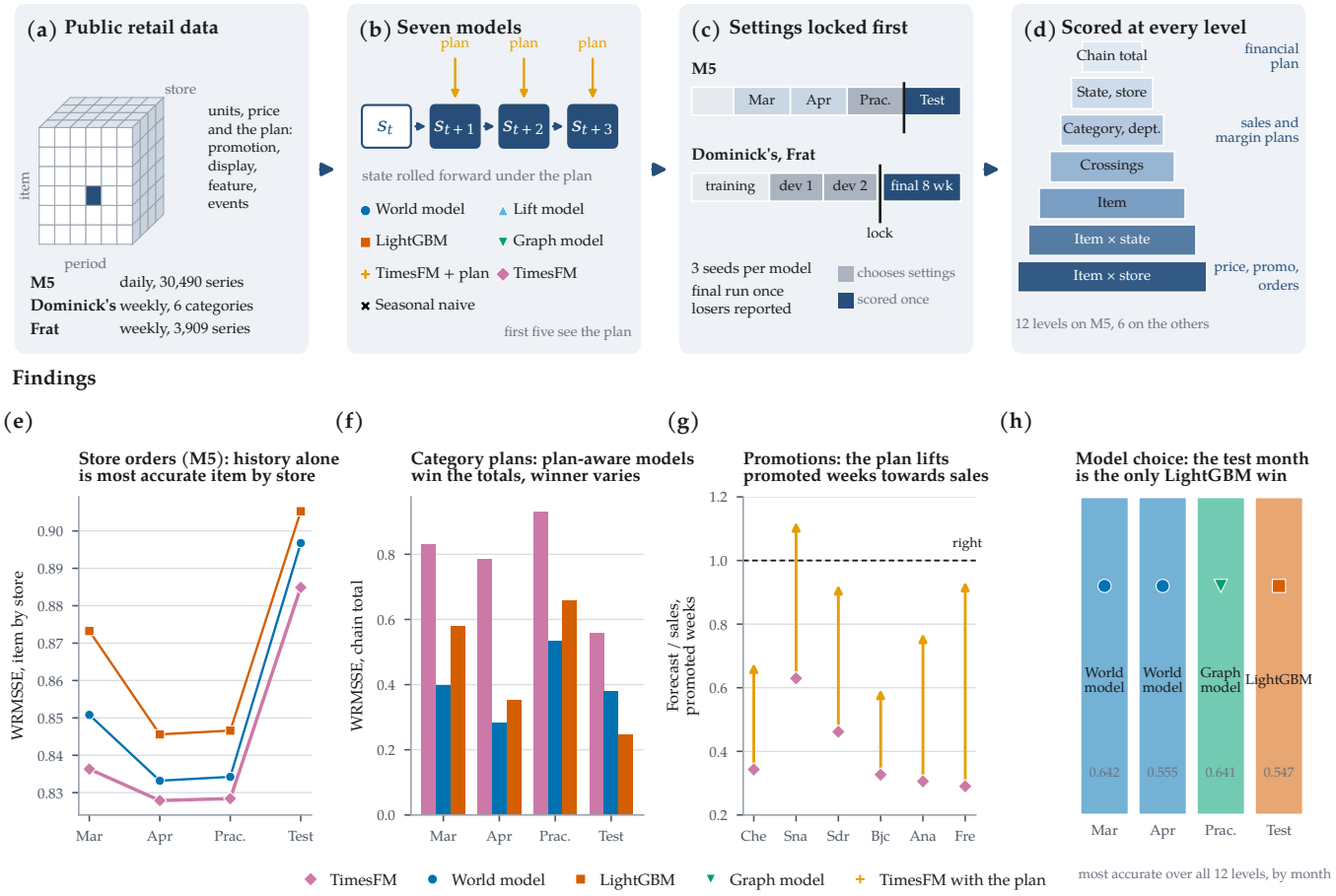


Figure 1: The study at a glance: which model is most accurate depends on the level the forecast serves, on how much of the sales the plan moves, and on the month. (**a–d**) Method. Three public data sets in one item, store and period table with the plan as known future inputs; seven models, five of which see the plan (the world model rolls each item's state forward under it); settings chosen on development windows and locked before the final windows; every model scored at every level of the hierarchy, from item by store, where orders and promotions are decided, to the chain total. TimesFM, TimesFM with the plan and the graph model were added after the final tests with settings fixed beforehand. (**e–h**) Findings, from the run records. (**e**) M5 item by store: TimesFM from history alone is lowest in all four months. (**f**) M5 chain total: the plan-aware models are far lower, and which one wins changes by month. (**g**) Dominick's promoted weeks: forecast against sales for TimesFM without the plan (diamond) and with it (arrow head). (**h**) The most accurate model over all 12 M5 levels in each month, with its WRMSSE.

sive LightGBM models fitted by store, category and department [4]. Our LightGBM follows that design without the ensembles. Neural global models include N-BEATS [29], DeepAR [30] and the Temporal Fusion Transformer [31], the last two taking known future inputs such as price. Forecasts from different models can be combined to reduce error [32, 33]; we test combinations in development. Graph neural networks share information between related series; our graph model uses GraphSAGE-style message passing [34], and our item-to-item experiments use attention [59] and conditional similarity networks [35], which learn several similarity spaces between items.

2.4 Pretrained foundation models

Pretrained time-series models forecast series they were not trained on from their history alone. TimesFM [36] is a decoder-only transformer pretrained on a large corpus of real and synthetic series; Chronos [38] tokenises values and trains a language model on them; Lag-Llama [39], Moirai [40] and TimeGPT [41] follow similar lines. TimesFM 2.5 adds a covariate method that fits a linear model on future inputs within the batch and forecasts the remainder with the pretrained model [37], which we use to give it the plan. We test on three data sets whether a pretrained model forecasting from history, with or without the plan, matches a model trained on the plan.

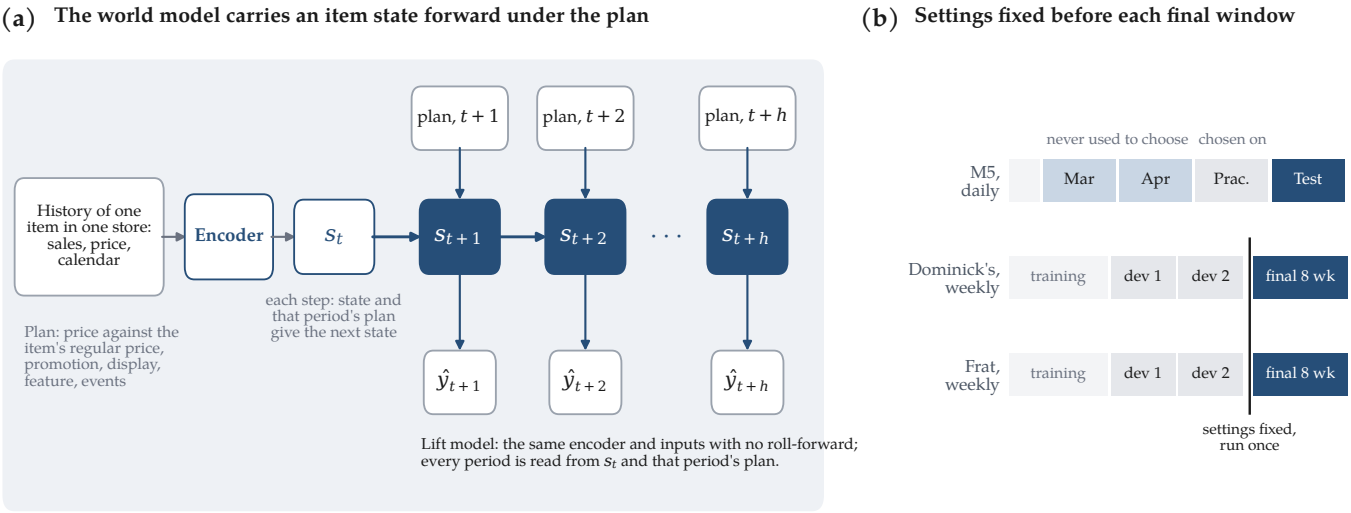


Figure 2: The world model carries each item's state forward under the plan, and every final window was run once with settings fixed beforehand. (**a**) An encoder turns the recent history of one item in one store into a state s_t. A transition step takes the state and one period's plan and gives the state one period later; each period's sales are read from the state as it stands after the plan so far. The lift model shares the encoder and inputs and reads every period from s_t. (**b**) On M5, settings were chosen on the practice window (25 April to 22 May 2016); the two earlier months were not used for any choice. On Dominick's and Frat, settings were chosen on two development windows before the final 8 weeks.

2.5 World models and latent prediction

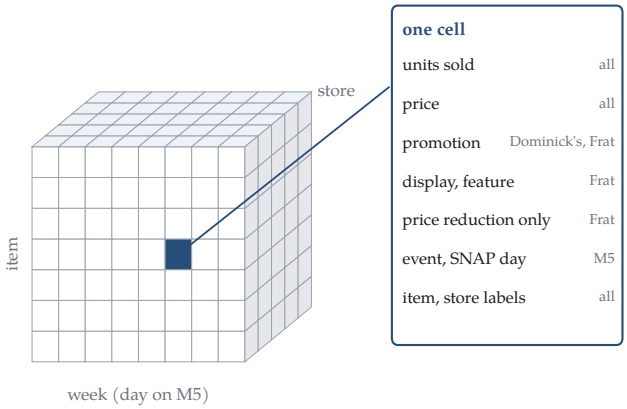
A world model learns a state of its environment and how that state moves under actions, so that the effect of a planned action can be rolled forward before it is taken [42, 43, 44]. LeCun [45] proposed joint-embedding predictive architectures (JEPA), which predict the representation of the future rather than the future itself; I-JEPA [46] and V-JEPA 2 [47] apply the idea to images and video, the latter with an action-conditioned predictor used for planning. Such models need a guard against a collapsed representation: a slowly updated copy of the encoder as the target [48], a variance and covariance penalty [49], or the single isotropic-Gaussian penalty of LeJEPA [50], also used in LeWorldModel [51]. TD-MPC2 [52] rolls a latent state forward under actions and weights near steps more in its loss. DWM [53] keeps the effect of an action apart from the world's own dynamics in the latent state, and AD-WM [54] trains world models to tell actions apart for counterfactual planning. Stable-worldmodel [55] provides a platform for evaluating world models under controlled changes, the same idea as our unusual-plan tests. Closest to us, WorldTS [56] learns covariate-conditioned latent dynamics for forecasting with known future inputs. Our earlier work, M-JEPA [57], compared JEPA fine-tuning recipes one by one with leakage-resistant splits; we carry that method into the development study. In retail the actions are the plan. Our world model applies the structure to sales with a recurrent transition [58] and layer normalisation [60] on top of a transformer encoder [59], and is compared with the same model without the roll-forward so that what carrying the state adds is measured on its own.

3 Data and the levels scored

All three data sets are public (Table 1) and share one table shape (Figure 3a). *M5* holds daily Walmart sales of 3,049 items in 10 stores across three states from 2011 to 2016, with prices by week, event days and the days on which SNAP food assistance is paid [1]. *Dominick's Finer Foods* holds weekly scanner data by store for a Chicago chain from 1989 to 1997, with price, a promotion code and cost [19]. We use six categories chosen before any final run (cheese, snack crackers, soft drinks, bottled juice, analgesics, frozen entrees) and report all 28 categories on the development windows (§A.2). *Breakfast at the Frat* holds weekly sales of 55 breakfast items in 77 stores from 2009 to 2012, with the shelf price, base price, and whether the item was on display, in the weekly circular (feature) or on a tag-only price reduction [20]. No raw data is redistributed.

Measure. We use the weighted root mean squared scaled error (WRMSSE) of M5 [1] at each level of the hierarchy: 12 levels on M5 (Figure 3b) and six on Dominick's and Frat. Each series' error is scaled by its own in-sample one-step error, and series are weighted by their dollar sales over the last weeks of training. We report the score over all levels (the M5 score), the chain total and item by store, the ratio of total forecast to total sales, and, on Dominick's and Frat, the ratio of forecast

(a) One row per item, store and period



(b) The levels a forecast is scored at, and who plans there

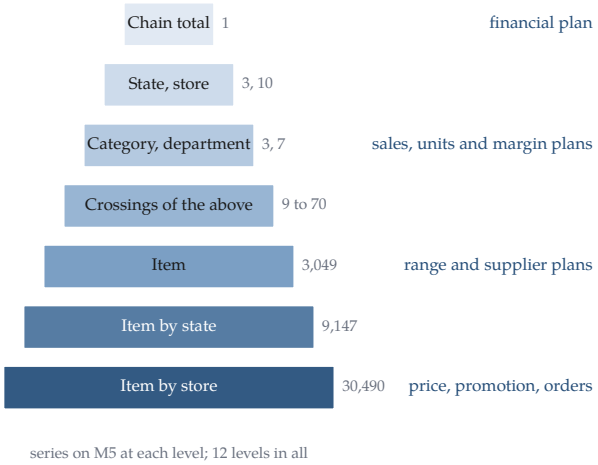


Figure 3: Every data set is one table of item, store and period, scored at every level a retailer plans at. (**a**) Each cell holds the units sold, the price and whatever plan the data set records; a lever a data set does not record is passed to the models as unknown, not as zero. (**b**) The M5 hierarchy with the number of series at each level, and the decisions usually made at each. Dominick's and Frat have six levels: total, category, store, category by store, item and item by store.

Table 1: Three public data sets, all scored item by store and at every level above it.

Data	Series	Plan inputs	Final windows
M5 (daily)	30,490	Price, event days, SNAP days	Four 28-day windows, 29 Feb to 19 Jun 2016
Dominick's (weekly)	Six categories	Price, promotion	Last 8 weeks of each
Frat (weekly)	3,909	Price, promotion, display, feature, price reduction only	Last 8 weeks, 16 Nov 2011 to 4 Jan 2012

to sales in promoted weeks, in the weeks just after a promotion ends and in ordinary weeks. Lower error is better.

4 Models

Seven models are compared in the final tests (Table 2). Five see the plan for the forecast periods and two do not.

Encoder and state. For item i in one store, an encoder (a transformer over the last L periods) reads sales and price, each scaled by the item's recent level ℓ_i, together with the calendar and the recorded levers, and gives a state s_t; the item and store labels enter the readout. Sales are scaled so that one model serves items selling one unit a month and items selling hundreds a day. The item's level is its average over the history window, or, on Dominick's, over its last 8 weeks (§A.2).

Table 2: Seven models: five see the plan, two forecast from history alone. Trained: fitted on the data set; the TimesFM variants are used as published.

Model	Plan	Trained	What it is
World model	yes	yes	Item state rolled forward under the plan
Lift model	yes	yes	Same encoder; baseline times plan lift; no roll-forward
LightGBM	yes	yes	One tree model per store, Tweedie loss
Graph model	yes	yes	LightGBM's inputs; message passing between related series
TimesFM with the plan	yes	no	TimesFM 2.5 with its covariate method
TimesFM	no	no	TimesFM 2.5, 512 periods of units
Seasonal naive	no	no	Sales one season earlier

World model. A gated recurrent transition f carries the state forward under the plan p_{t+h} for each future period,

$$s_{t+h}(p) = \mathrm{LN}(f(p_{t+h},\, s_{t+h-1}(p))), \quad s_t(p) = s_t, \qquad (1)$$

where LN is layer normalisation (used on M5 and Dominick's; §A). The plan holds the period's price as a share of the item's regular price (its highest price in the last 12 periods) and the recorded levers. Expected sales are a baseline under the "nothing planned" plan p^0 (regular price, no lever) times a lift,

$$\begin{aligned}\log \mu_{i,t+h} = {} & \log \ell_i + b(s_{t+h}(p^0), p^0_{t+h}) \\ & + g(s_{t+h}(p), p_{t+h}) - g(s_{t+h}(p^0), p^0_{t+h}),\end{aligned} \qquad (2)$$

Table 3: The world model as run in the final tests. Read from the run manifests in `results/promoted/`. Periods are days on M5 and weeks on Dominick's and Frat.

Setting	M5	Dominick's	Frat
Periods of history read	112	26	26
Periods of training data	730	156	90
State size	128	64	64
Encoder layers	3	2	2
Attention heads	4	4	4
Training steps	30000	6000	3000
Batch size	1024	512	256
Learning rate	0.001	0.001	0.001
Normalised roll-forward	yes	yes	no
Level from the last 8 weeks	no	yes	no
Brand switching	no	yes	no
Fits averaged per run	1	3	1

with b and g small networks that also take the item's labels and the horizon step, and the log forecast bounded to $[-10, 10]$. The lift is exactly zero when nothing is planned, and the baseline does not move when the plan changes; both properties are checked by tests in the repository. Training minimises a Tweedie loss on sales relative to ℓ_i, with batches of item, store and date drawn at random.

Brand switching (Dominick's). Items forecast one by one each add their own promotion lift, but in a store and week where several items are promoted much of each item's lift is taken from the others. Following attraction models [16, 17], the world model fits on the last 8 training weeks one number per run, the share α of the summed lift that the category total keeps, and in each store and week rescales its item forecasts so that the category total is the baseline total plus α times the summed lift. α is the split that makes category totals fit best: a forecast-implied allocation of lift, not an identified estimate of how much promoted sales come from other items, because promotions are not assigned at random. Each item keeps its share of the total as forecast. On Dominick's three fits with different seeds are averaged in each run. Table 3 gives the settings of the final runs.

Lift model. The same model without the roll-forward: $s_{t+h}(p) = s_t$ for every h. The two share inputs and training, so the difference between them is the roll-forward. On Frat its readout uses dropout [61] and decoupled weight decay [62], which made it steadier across seeds in development.

Comparison models. *LightGBM* [27]: one direct model per store, Tweedie loss, sales at lags no shorter than the horizon, rolling averages and spread of sales, price against the item's highest recent price, calendar and plan inputs, 800 boosting rounds on M5. *Graph model*: LightGBM's inputs, with one round of GraphSAGE-style message passing [34] between the same item in other stores and items of the same department in the same store, Tweedie loss, 6,000 steps. *TimesFM* [36, 37]: the 200M-parameter TimesFM 2.5 checkpoint given each series' last 512 periods of units; we use its mean output, because its default median forecasts 27% to 31% below M5 totals (§7). *TimesFM with the plan*: the same model through its covariate method, a linear model on the plan inputs fitted over groups of 2,048 series, then TimesFM on what that model leaves. *Seasonal naive*: sales one season earlier (a week on M5).

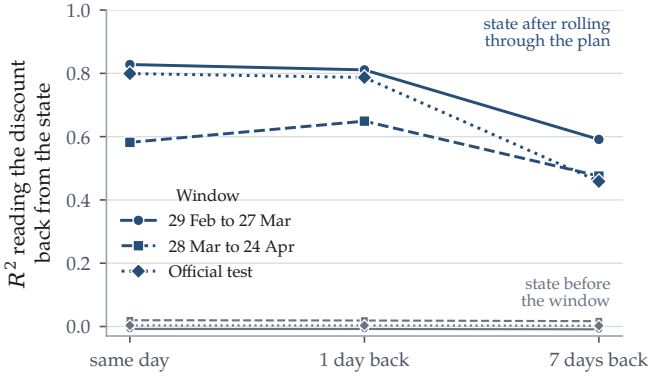


Figure 4: The rolled state holds the plan, and still holds most of it a week later. R^2 of a ridge read-out of the day's discount (price against regular price) from the frozen world model's state, fitted on half of 6,000 item-store pairs and scored on the other half. Blue: the state after rolling through the window's plan. Grey: the state before the window, which has not seen the plan. Seed 1 in three M5 windows.

What the state holds. A probe on the frozen world model shows that the carried state takes in the plan (Figure 4). The day's discount can be read back from the rolled state with R^2 0.58 to 0.83 across three M5 windows, against -0.01 to 0.02 from the state before the window; event and SNAP days at 0.98 to 0.99; and the discount still at 0.46 to 0.59 seven days later. This shows that the plan is in the state; it does not show that the forecast uses it well, which the results test.

5 Protocol

Development and final windows. On Dominick's and Frat, the last 8 weeks of each data set are the final window, touched once. Development used the two 8-week windows before it, each forecast from everything earlier (Figure 2b). On M5, development used the practice window, 25 April to 22 May 2016, the 28 days before the official test month. After the final tests, two earlier M5 months, 29 February to 27 March and 28 March to 24

April 2016, were run with settings unchanged; they were not used for any choice. One exception is recorded: on Frat the three benchmarks of §A.1 were first scored on three windows that include the final 8 weeks, before that window was locked; none was tuned afterwards.

Rules fixed in advance. A component of our model was kept only if removing it made the development score worse; a change that doubles the training cost was kept only if it lowered the score by more than the seed-to-seed range. A benchmark received every input our model received. Each final test ran three seeds per trained model, and the range is reported. The final tests and their settings were written down on 9 October 2026 before any of them ran, as copies of the development configurations with only the name and the scored window changed, which a test in the repository checks. Each was run once and is reported as it came out.

Models added after the final tests. TimesFM, TimesFM with the plan and the graph model were added after the final tests as comparisons, with settings fixed before their first run and no development windows of their own.

Development study. How every setting of the tested models was chosen, the changes that were kept and those that were not, and the development scores on all three data sets are in the supplement (Section A), so that the main text holds the hypotheses and the final evidence.

6 Final results

The results follow the order of the contributions: the level at which a model is judged (§6.1), the number of months it is judged on (§6.2), the plan (§6.3, §6.4), and what carried over from development (§6.5). Table 4 and Table 5 hold every final score.

6.1 The most accurate model depends on the level

On M5, TimesFM given only each item's sales history is the most accurate model item by store in all four months (Figure 5): 0.836, 0.828, 0.828 and 0.885, against 0.851, 0.833, 0.834 and 0.897 for the world model and 0.873, 0.846, 0.847 and 0.905 for LightGBM. Among the trained models, the world model and the lift model are ahead of LightGBM and the graph model item by store in every month, and in both confirmation months every world-model seed is ahead of every LightGBM seed.

At the total the order reverses. TimesFM forecasts 5% to 10% below total sales in every month, so its total error is higher than seasonal naive's in three of the four months (0.830 against 0.658 in March) and level with it on the test month (0.558 against 0.560). The world model has the lowest total error in March (0.398) and April (0.283), the graph model on the practice window (0.455) and LightGBM on the test month (0.247).

Figure 6 shows where the order changes. From the chain total down to store by department a plan-aware model is the most accurate in every month, with the world model's error typically 20% to 50% below TimesFM's. At the three item levels (item, item by state, item by store) the models are within a few percent of each other, and TimesFM is the most accurate in three of the four months, item by store in all four. A model that is best for store orders on M5 is therefore not the model to use for a category plan. The combined M5 score weights the twelve levels equally, and nine of them are above the item, so a ranking on it mostly reflects the upper levels.

6.2 One test month misleads

Over all twelve levels, the official M5 test month (23 May to 19 June 2016) is the only one of the four in which LightGBM is the most accurate model: 0.547 against 0.638 for the world model, level with the 10th-ranked entry of the competition (0.547), with the winner at 0.520 and the best statistical benchmark at 0.671. In March the world model is ahead, 0.642 (seeds 0.632 to 0.648) against 0.709 (0.706 to 0.713); in April the two are level, 0.555 against 0.557; and on the practice window the graph model is lowest (0.641), with the world model at 0.656 and LightGBM at 0.715.

The ranking follows the level of the total forecast (Figure 7a): across models, months and seeds, the size of the total bias and the total error have a correlation of 0.87. Between the world model and LightGBM, LightGBM forecasts 3% to 6% below total sales in March and April and 2% below on the test month; the world model is within 1% in March and April and 2.6% above on the test month. In each of these three months the one of the two whose total is closer to sales has the lower total error and the lower error over all levels. Grouping April's item-store pairs by their March sales rate (Figure 7b), both models are within 6% of sales in every group that sold in March; LightGBM is under in every group and the world model is closer on 50.6% to 54.2% of pairs in each. The world model's weak spot is pairs that sold nothing in March, where it forecasts 62% above April sales.

A comparison on the official test month alone would have ranked LightGBM first. On three of the four months it is not.

Table 4: On M5 the best model depends on the level scored and on the month. WRMSSE on the four 28-day windows (lower is better), mean over seeds; seeds per window in the second column. All: the 12 levels together; Total: the chain total; I×S: item by store. The lowest value in each column is in bold. TimesFM with the plan has no recorded total. The lift model's April mean includes one failed seed (2.603 overall; §7).

		29 Feb to 27 Mar			28 Mar to 24 Apr			25 Apr to 22 May (practice)			23 May to 19 Jun (test)		
Model	**Seeds**	All	Total	I×S	All	Total	I×S	All	Total	I×S	All	Total	I×S
World model	3/3/3/3	**0.642**	**0.398**	0.851	**0.555**	**0.283**	0.833	0.656	0.535	0.834	0.638	0.379	0.897
Lift model	3/3/3/3	0.664	0.427	0.846	1.246	0.964	0.834	0.713	0.623	0.832	0.663	0.368	0.896
LightGBM	3/3/1/3	0.709	0.578	0.873	0.557	0.351	0.846	0.715	0.659	0.847	**0.547**	**0.247**	0.905
Graph model	3/3/3/3	0.778	0.621	0.878	0.644	0.493	0.845	**0.641**	**0.455**	0.851	0.673	0.412	0.909
TimesFM	1/1/1/1	0.854	0.830	**0.836**	0.819	0.786	**0.828**	0.896	0.930	**0.828**	0.729	0.558	0.885
TimesFM with the plan	1/1/1/1	0.858	–	0.837	0.772	–	0.829	0.929	–	0.830	0.719	–	**0.883**
Seasonal naive	1/1/1/1	0.889	0.658	1.121	0.923	0.668	1.126	0.870	0.751	1.100	0.847	0.560	1.176

Table 5: On the promotion-driven weekly data, models that see the plan lead, and which one leads changes by category. WRMSSE on the final 8 weeks (lower is better). World model, lift model and LightGBM: mean of three seeds, settings fixed on the development windows. TimesFM and TimesFM with the plan: one run each, added after the final tests. The lift model was run on Frat only. The lowest value in each row and group is in bold.

	All levels					Item by store				
Data	World	Lift	LightGBM	TimesFM	TimesFM+plan	World	Lift	LightGBM	TimesFM	TimesFM+plan
Frat	0.681	0.714	**0.549**	0.965	0.686	0.665	0.677	**0.608**	0.827	0.641
Dominick's										
Cheese	0.897	–	0.708	0.829	**0.545**	0.715	–	0.702	0.792	**0.699**
Snack crackers	0.448	–	**0.412**	0.468	0.475	0.608	–	**0.596**	0.640	0.614
Soft drinks	**0.830**	–	0.989	1.070	0.974	0.734	–	**0.652**	0.895	0.780
Bottled juice	0.987	–	0.994	**0.939**	0.990	0.816	–	**0.750**	0.782	0.754
Analgesics	**0.867**	–	0.978	1.008	0.889	**0.713**	–	0.714	0.790	0.742
Frozen entrees	0.485	–	**0.478**	0.563	0.566	0.557	–	**0.529**	0.682	0.599

6.3 How much the plan matters

The plan matters in proportion to how much of the sales it moves. On M5, where prices change rarely, giving TimesFM the plan changes its item-by-store error by at most 0.002 in any month (0.837 against 0.836 in March). On Dominick's and Frat, where promotions drive the weeks, it changes a great deal (Figure 8).

Without the plan, TimesFM is behind both the world model and LightGBM item by store on Frat and in five of six Dominick's categories; in bottled juice it is between them. This is the reverse of M5. The reason shows in the promoted weeks (Figure 9a): TimesFM forecasts 29% to 63% of what sold in a promoted week, because nothing in its input says the promotion is coming.

Given the plan, TimesFM forecasts 60% to 113% of promoted-week sales, and its overall error improves on Frat (0.965 to 0.686) and in three of six categories (cheese, soft drinks, analgesics). It becomes the most accurate model in cheese (0.545 against 0.708 for LightGBM) and is level with the world model on Frat (0.686 against 0.681), but stays behind a model trained on the plan in the other categories. In bottled juice the plan made it slightly worse (0.939 to 0.990).

The trained plan-aware models also fall short in promoted weeks: the world model forecasts 48% to 88% of promoted-week sales and LightGBM 50% to 84%. Among the trained models, which one is more accurate depends on the category (Table 5): over all levels the world model is ahead in soft drinks (0.830 against 0.989) and analgesics (0.867 against 0.978), with every world-model seed ahead of every LightGBM seed in both; level in bottled juice; slightly behind in frozen entrees (0.485 against 0.478); and behind in cheese, snack crackers and on Frat. Item by store LightGBM is level or ahead in every category. On Dominick's the world model's gains are at the totals, the opposite pattern to M5.

6.4 First-time price cuts

In the April M5 window, HOUSEHOLD_1_146 had sold almost nothing for a year at 5.98, then sold 120 units

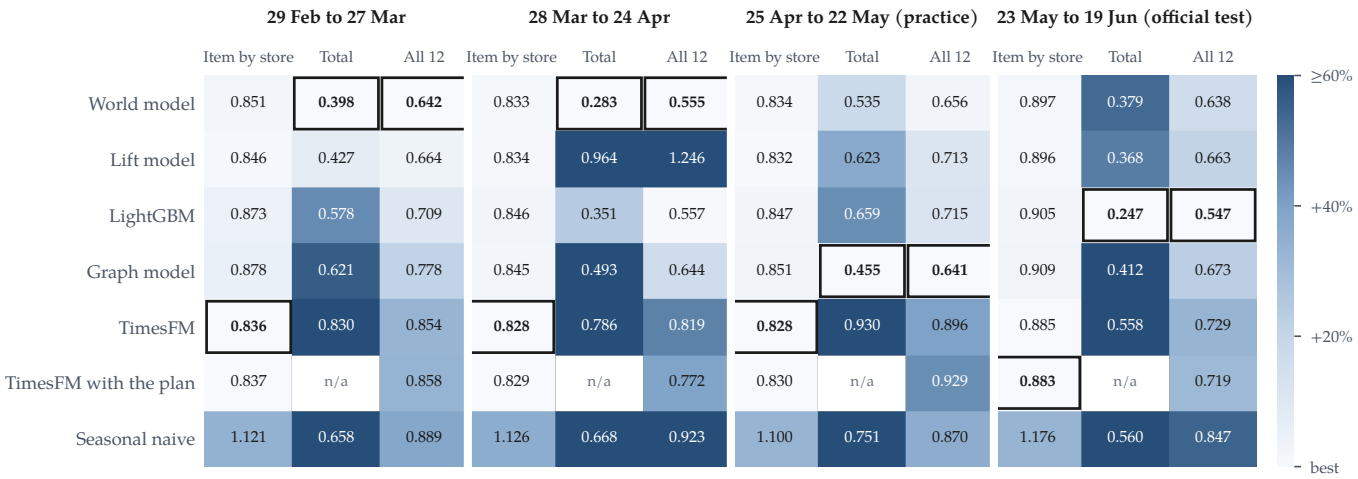


Figure 5: On M5, TimesFM from history alone has the lowest item-by-store error in every month; at the total and over all levels the plan-aware models do, and which one changes by month. WRMSSE of every model, four windows, three grains; cell shade: how far above the lowest value in its column (white: the lowest, outlined). Trained models: mean of three seeds (LightGBM on the practice window: one run). The lift model's April mean includes one failed seed (§7). TimesFM with the plan has no recorded total.

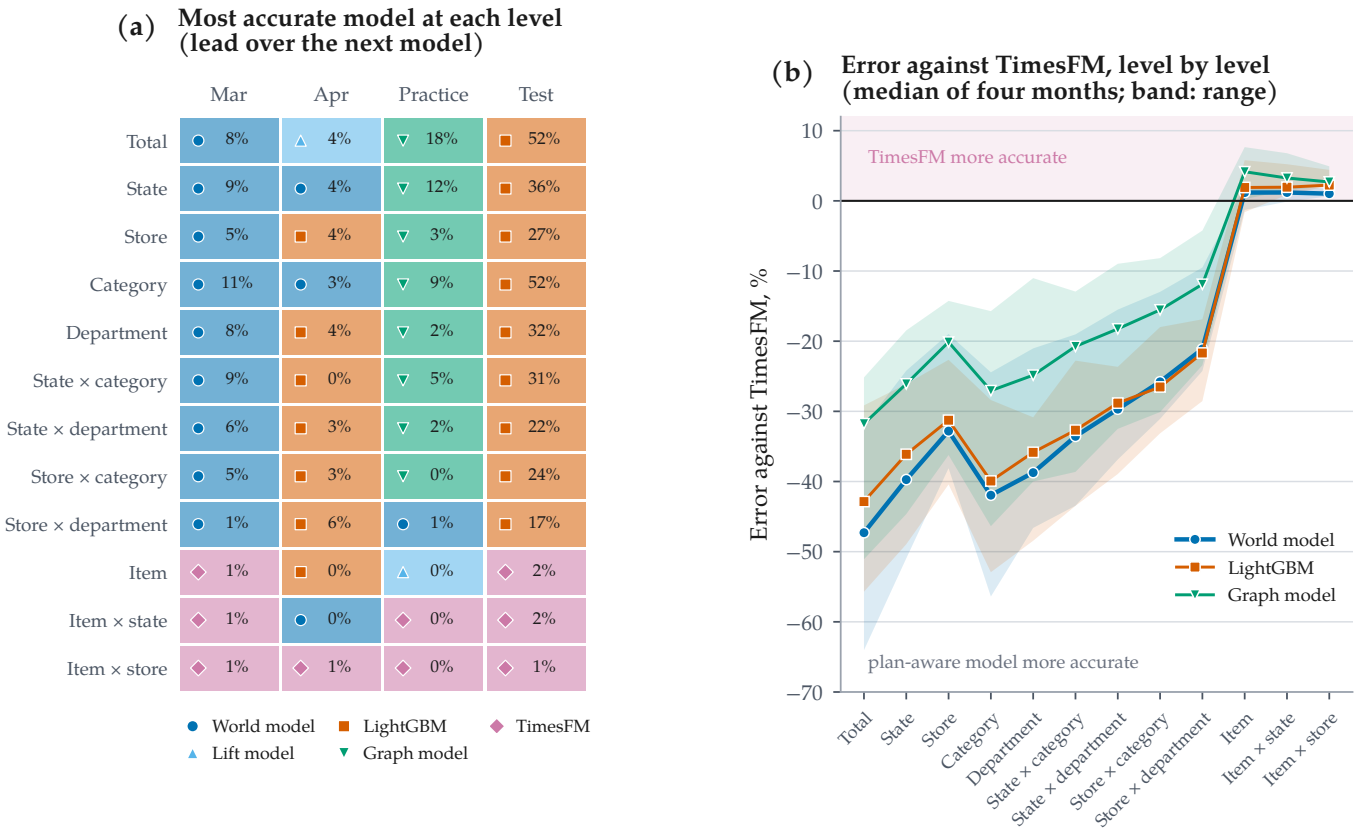


Figure 6: From the chain total down to store by department a plan-aware model is most accurate; at the three item levels the models are within a few percent and TimesFM usually leads. **(a)** The most accurate of five models at each M5 level in each window (cell colour and marker), with its lead over the next model in percent (lift model: median of its seeds). **(b)** Error of the world model, LightGBM and the graph model against TimesFM at each level: median of the four windows (line) and range (band); below zero the plan-aware model is more accurate.

in four stores over a window that held a one-week cut to 4.50 (Figure 9b). Models working from history forecast almost nothing: LightGBM 3.0, 3.0, 3.2 units across seeds, TimesFM 7.8. The world model, which reads the cut through its plan, forecast 121, 600, 243; the graph model 59, 45, 55; the lift model 79, 206,886, 316.

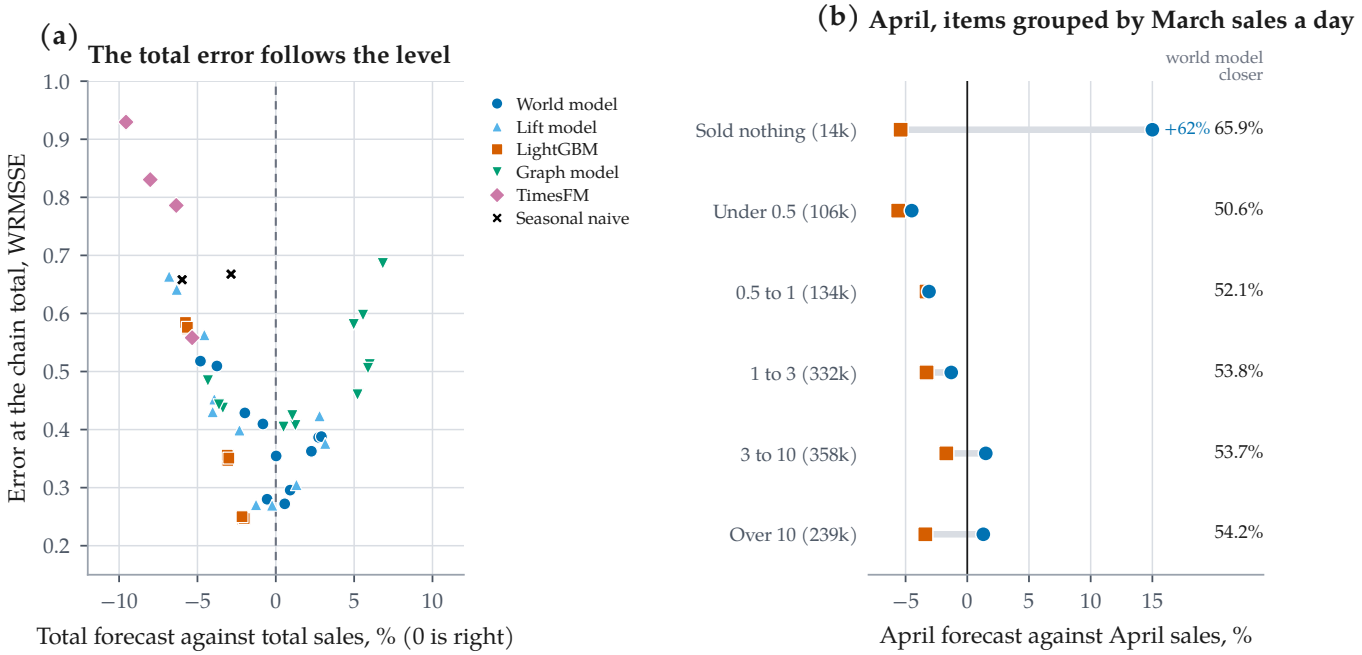


Figure 7: Whichever model gets the month's total closer has the lower total error, and that changes by month. (**a**) Error at the chain total against the total forecast's bias, one point per model, M5 month and seed ($n = 50$; one failed lift seed lies off the chart). The correlation of the size of the bias with the error is 0.87. Seasonal naive's bias was not recorded on the two later windows. (**b**) April, with each item-store pair grouped by its average daily sales in March (units sold in April in brackets), so that the grouping does not depend on the sales being scored: forecast against sales for the world model (circle) and LightGBM (square), and the share of pairs where the world model is closer.

Models that read the plan respond to a price change that history has not shown, and LightGBM, which sees price only against its own recent highest, responds little. How large the response should be is not fixed by the data: the world-model seeds differ by a factor of five, and one lift-model seed forecast over 200,000 units. For the same window, FOODS_3_443 in one store had a 68% cut and sold 4 units; every plan-reading model forecast more (the world model up to 26).

6.5 What carried over from development

Development rankings carried to the final weeks in some places and not in others. On Dominick's the world model was ahead of LightGBM in development in analgesics, frozen entrees and soft drinks; in the final weeks it was ahead in analgesics and soft drinks, by more than the seed ranges, and slightly behind in frozen entrees. On M5 the practice-window lead (0.656 against 0.715) held in March, was a tie in April and reversed on the test month. On Frat the three trained models were level in development (0.406 to 0.421) and LightGBM was clearly ahead in the final weeks (§7). In each case the difference between development and final followed what changed in the data: the month's level on M5, the holiday weeks on Frat. A development win is evidence about the windows it was measured on, and the paper reports both.

7 Negative and degenerate results

We report the failures at the same length as the results. Two had a measured cause; the other changes and comparison models are in Table 6.

A lift-model seed that broke on five series. In April, lift-model seed 2 scored 2.603 over all levels against 0.566 and 0.570 for seeds 1 and 3, and forecast 1,413,724 units against 1,183,626 sold. Five item-store series hold all of the excess. Each had its first price cut in a year inside the window (the four HOUSEHOLD_1_146 stores of §6.4, and FOODS_3_443 in one store, 2.50 to 0.79 for three weeks). Seed 2 read the cuts as lifts up to the model's bound, about 22,000 times the item's level per day. Item by store, weighted by dollar sales, the failure is invisible (0.834); at the totals it dominates. A price planner looking at items would see a few absurd forecasts; a category planner would see the month's total 19% too high. The world model, with the same readout and the roll-forward, did not fail on these series in any seed.

Frat holiday weeks. On the Frat development windows the world model, lift model and LightGBM were level (0.406 to 0.421). On the final 8 weeks the world model fell to 0.681 against 0.549 for LightGBM, and every LightGBM seed beats every world-model seed. The final weeks are the holiday period, which the development

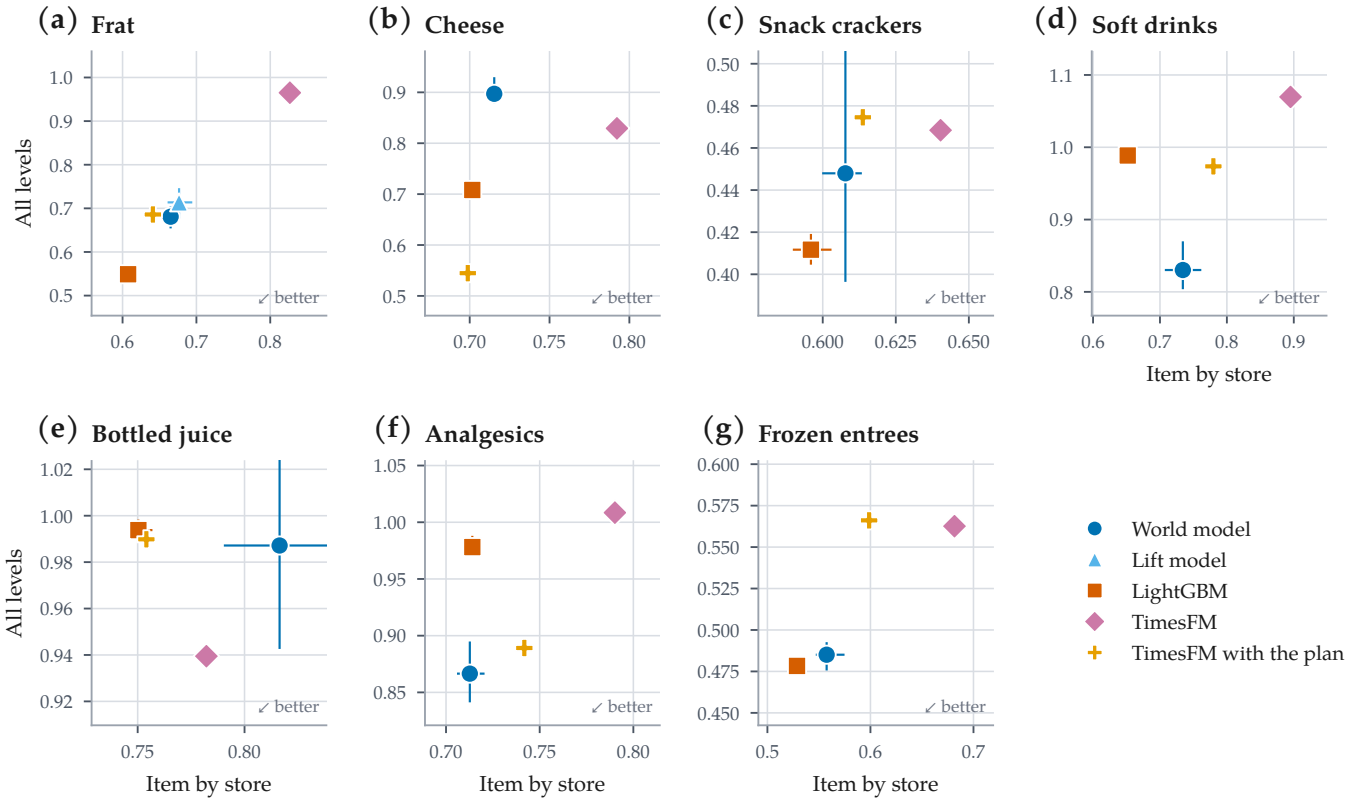


Figure 8: On the weekly data each data set has its own trade-off between item-by-store and all-level accuracy, and TimesFM without the plan is behind both trained models item by store in six of seven. Final 8 weeks: item-by-store WRMSSE (horizontal) against all-level WRMSSE (vertical), lower-left is better. World model, LightGBM and (Frat only) lift model: mean and seed range in both directions; TimesFM variants: one run. Axes differ by panel.

windows (late July to mid November) did not contain (Figure 10). The world model's 8-week total is right (forecast against sales 1.000, LightGBM 0.937), but it is 15% to 17% under in the two December peak weeks and 24% over in the week after Christmas. LightGBM, with week-of-year and month inputs and training data that include the previous holiday season, follows that shape more closely. The world model is still ahead of the lift model here (0.681 against 0.714).

TimesFM's default output. TimesFM's default point forecast is the median of each day's sales. For an item that sells nothing on most days the median is near zero while the mean is not, so the median forecast is 27% to 31% below M5 totals in every month, and its upper-level error is worse than seasonal naive's (overall 1.71 to 1.88). Item by store it was close to LightGBM. A median is the wrong target for a merchant's units and dollars at category level; the mean output, used throughout, does not have the problem.

A withdrawn claim. An earlier reading that the world model was more accurate than LightGBM on fast-selling items came from grouping pairs by their sales in the scored window, which pulls forecasts in high groups towards under and in low groups towards over for both models. Grouped by an earlier window (Figure 7b), the advantage is small and the same in every group. We do not make the claim.

Changes that did not become defaults. Table 6 lists every change and comparison model that lost, and §A gives the development detail. Longer training, weekday and event inputs, a long-run level, a totals term in the loss, a count likelihood, a separate plan path in the state, feeding the model's own forecast into the roll-forward and item role labels either made no difference beyond seed noise or made the model worse. The graph model is level with LightGBM item by store, whose inputs it shares, so sharing information between related series added nothing on these inputs; neither did item-to-item links in our own model on Frat or Dominick's.

8 Discussion

A retailer should choose a forecasting model by the decision it feeds, scored at that decision's level and over several periods (Table 7).

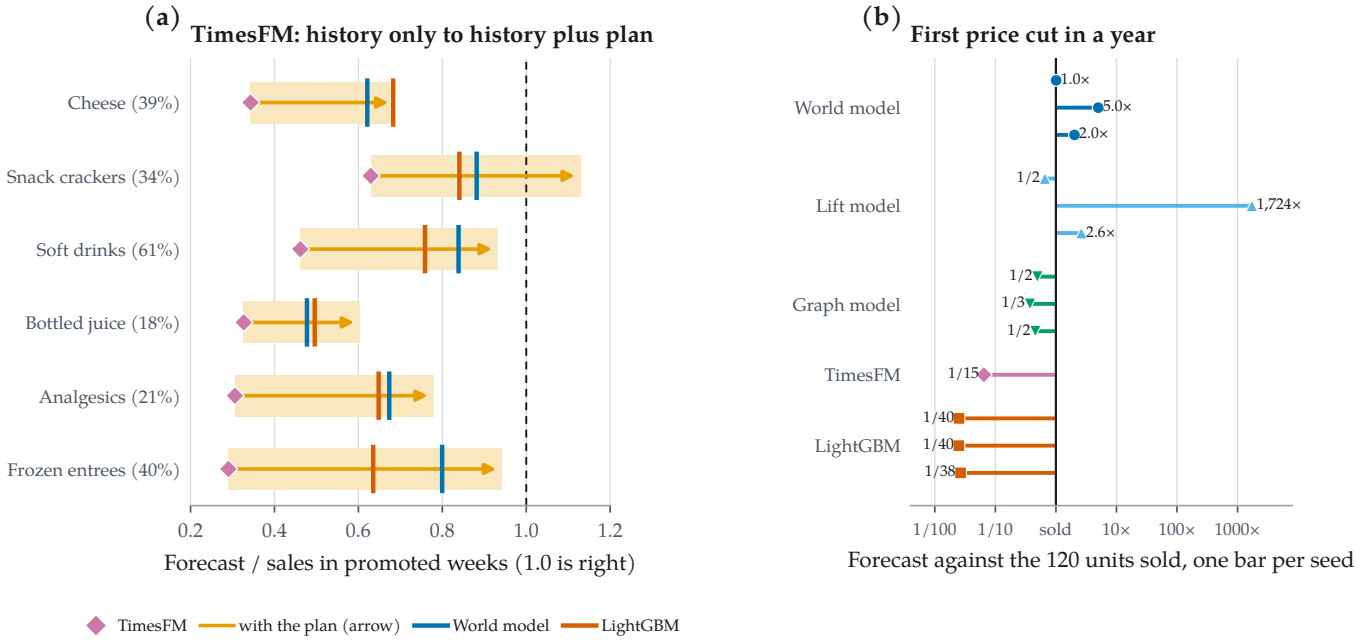


Figure 9: Given the plan, TimesFM closes most of its promoted-week shortfall; for a price cut history has not shown, plan-aware models respond and history-only models do not. (**a**) Forecast against sales in the promoted weeks of the final 8 weeks, by Dominick's category (promoted weeks' share of the category's units in brackets): TimesFM from history alone (diamond) to TimesFM with the plan (arrow head), with the world model and LightGBM (ticks). (**b**) HOUSEHOLD_1_146 in the four M5 stores where it had its first price cut in a year (5.98 to 4.50 for one week) inside the April window: each seed's forecast over the 28 days against the 120 units sold, log scale.

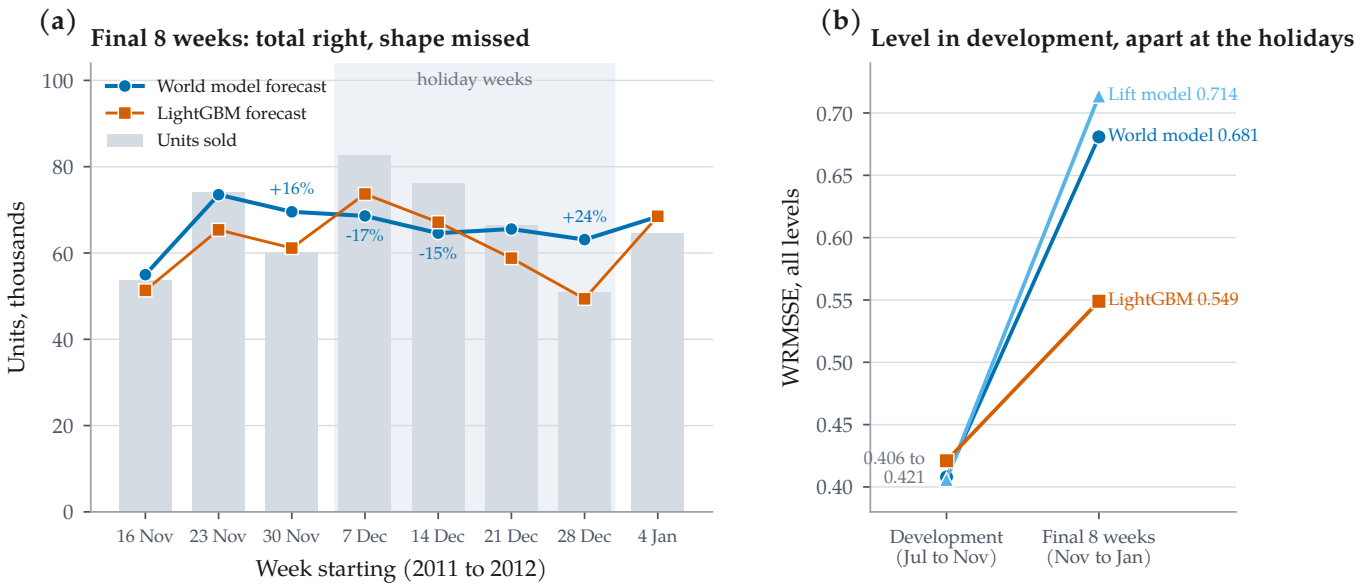


Figure 10: On Frat the world model gets the 8-week total right and misses the holiday shape, which the development windows did not contain. (**a**) Units sold each week of the final window over all item-stores (bars), and the weekly forecasts of the world model and LightGBM, three seeds averaged; labels: the world model's largest weekly misses. (**b**) All-level WRMSSE on the development windows and on the final 8 weeks.

Why the level changes the answer. Item-by-store error is dominated by the noise of single items selling a few units, which history read well can match and the plan adds little to when prices rarely move. The totals are dominated by the month's level: a 2% bias in the same direction on every item is invisible item by store and decides the category total. TimesFM forecasts from history what an item usually sells and has no input that says this month will differ; the plan-aware models are trained across all items and stores and see events, SNAP days and prices, and get the level closer. On weekly promotion-driven data the plan moves the single items

Table 6: Every change and comparison model that did not become a default. Overall WRMSSE unless stated; "against" gives the default it was compared with on the same window. dev.: development windows before the final 8 weeks. Practice: the M5 window 25 April to 22 May 2016. Each row is a section of `results/README.md` with its run folders.

Change or model	Where tested	Result
Twice the training steps	Dominick's (6) and Frat, dev., 3 seeds	Better in 2, worse in 3, level or mixed in 2
Weekday as seven inputs	M5 practice, seed 1	0.653 against 0.646
Event kinds (with weekday)	M5 practice, seed 1	0.646 against 0.646
60,000 training steps	M5 practice, seed 1	0.642 against 0.646; gain below seed range 0.029
Item level against past-year level	M5 practice, seed 1	0.680 against 0.646
Totals term in the loss	Dominick's analgesics, dev.	0.819 against 0.683; total 8% under
Negative binomial likelihood	Frat, analgesics, dev.	Level
Plan split, latent check, near-week weighting	Frat, dev., 5 seeds	0.417 to 0.427 against 0.408
Own forecast fed into the roll-forward	Dominick's (6), dev.	Worse overall in 4 of 6
Item role labels	Dominick's (6), dev.	No effect on LightGBM; snack crackers 2.422 for the world model
Averaging three fits	M5 practice, seed 1	0.665 against 0.656 (kept on Dominick's)
Graph model over product attributes	M5, four windows, 3 seeds	Level with LightGBM item by store
TimesFM, median output	M5, four windows	Overall 1.71 to 1.88; totals 27% to 31% under
Lift model (no roll-forward)	M5, four windows, 3 seeds	One seed failed in April (2.603)

Table 7: Each decision has its own level, and the results point to a different model for each.

Decision	Level	What the results say
Store orders and inventory	Item by store, day or week	Where prices rarely move (M5), the world model is more accurate than LightGBM in all four months and TimesFM, from history alone, slightly more accurate than both. Where promotions drive the weeks (Dominick's, Frat), LightGBM is more accurate than the world model
Promotion, price change, supplier funding	Item by week, under a planned change	A model that reads the plan is needed; history alone forecasts 29% to 63% of promoted-week sales and does not respond to a first price cut. The size of a response to a plan unlike history needs a bound
Category sales, units and margin plans	Category or chain, week or month	Not TimesFM (5% to 10% short). Between the world model and LightGBM the winner changes by month and category (M5 totals: world model two months, LightGBM one, the official test month; Dominick's: world model two categories, LightGBM three and Frat), so compare over several periods
Promotion planning across a category	Category by store-week	Forecast the category total, not a sum of item lifts: in snack crackers the summed item lifts overstated the total, and the fitted split kept under two thirds of them
Choosing a model	Any	Score at least three or four periods; one period reversed the ranking on M5

too, so the plan-aware models win item by store as well.

What the roll-forward adds. Against the lift model, which differs only in the roll-forward, the world model is level or better over all levels in every M5 month and on Frat, it was steadier across seeds once the rolled state was normalised, and it did not fail on the first-time price cuts that broke a lift-model seed. Item by store it is not more accurate than the lift model, and on M5 neither is more accurate than a pretrained model that sees no plan at all. The development tests designed to show what rolling a plan forward buys (plan A against plan B, unusual plans) did not separate it from LightGBM. On these data, the case for the roll-forward is steadiness and a bounded response to unfamiliar plans, not accuracy on familiar ones.

Two cautions. A direct lift readout can extrapolate a price cut it has never seen far beyond any sale, and whether it does depends on the seed; a plan-driven forecast needs a bound and a check on series whose plan falls outside their own history. And a forecast that is accurate item by store can be several percent off the category total, which is the number a merchant plans against.

9 Limitations and threats to validity

Windows and seeds. M5 has four final windows; Dominick's and Frat one each. On the M5 practice window LightGBM was run once. Much of the development study used one seed, and single-seed world-model runs varied by up to 0.07 on analgesics; the decisions that mattered were confirmed on three to five seeds. Three seeds show the spread of training, not of the data.

Comparison models added later. TimesFM, TimesFM with the plan and the graph model were added after the final tests, with settings fixed before their first run but without development windows of their own. Each TimesFM variant is one run. TimesFM with the plan fits its linear plan model within groups of 2,048 series, a memory limit, so the fit is shared within each group.

What lift means. Merchants promote in weeks they expect to sell well, so part of a measured lift is good timing. The lift is what the forecast attributes to the plan, not what the plan caused. Our companion audit [63] measures the gap: in generated store data where merchants plan into strong weeks, the world model's and LightGBM's lifts were a third too large while every check against the model, the sales and each other passed, and the world model answered a question about a lasting regular-price change with the response to a temporary cut. On Dominick's, zone price deviations give a measured but imprecise response to regular prices. Plan responses in this paper are forecasts, not causal estimates.

The size of a plan response. The world model's response to a first-time price cut varied by a factor of five across seeds. The data do not identify it, and a deployed system would need a bound or a prior.

What the data do not record. None of the data sets records stock, so a sell-out looks like low demand; or shelf space, or placement within a display or circular. Weather is not known 8 weeks ahead. The data were collected between 1989 and 2016, and their plan inputs differ, so the results compare models within each data set, not across them.

10 Conclusion

Which forecasting model is most accurate depends on the level it is judged at, the month, and how much of the sales the plan moves. On M5 a pretrained model reading history alone is the most accurate item by store in every month, while plan-aware models are more accurate at the totals, and the official test month is the only one of four in which LightGBM wins overall. On promotion-driven data a model without the plan forecasts well under half of promoted-week sales in most categories, and adding the plan to a pretrained model recovers much of that without replacing a model trained on the plan. A world model that carries each item's state forward under the plan responds to price changes history has not shown, in the right direction and with an uncertain size, and learned from store sales how much of a promotion's lift is new to a category. Retail forecasts should be chosen and reported by the decision they feed, at that decision's level, over several periods, with the losing models stated.

11 Reproducibility and compute

Records. All code, configuration files and tests are in the repository. A run refuses to start on uncommitted code, and writes a manifest holding the code commit, the full configuration and its checksum, the seed, checksums of every data file read, Python and package versions, the machine (processor, cores, memory, GPU and its memory, CUDA version) and, from 9 October 2026 on, its start and finish times and the seconds spent fitting and forecasting each window. The manifest and scores of the 110 runs read by this manuscript are committed under `results/promoted/`; its figures, tables and quoted final-test numbers are generated from them by `paper/figs/make_figures.py` and `paper/figs/make_tables.py` (`make` in `paper/` rebuilds the manuscript). Numbers from runs not in that folder (the development study, TimesFM on Dominick's and Frat, TimesFM with the plan, and the analyses of saved forecasts) are copied into `paper/figs/rwdata.py` from `results/README.md`, which names each run folder. Forecast files and model checkpoints are kept outside the repository, with their checksums recorded in each run's scores so a stored file can be shown to be the one the run produced.

Seeds and repeat runs. Trained models in the final tests ran with seeds 1, 2 and 3. Runs on a GPU do not repeat exactly; the seed ranges in every table show how far apart repeats fall: on M5, LightGBM seeds differ by 0.001 to 0.012 over all levels and world-model seeds by 0.014 to 0.029. TimesFM has no training randomness and was run once per window.

Compute. Every run used one Google Colab machine: an NVIDIA Tesla T4 with 14.6 GB of memory, a 2-core Intel Xeon CPU @ 2.00GHz and 12.7 GB of memory, PyTorch 2.11.0+cu130 with CUDA 13.0. LightGBM ran on the machine's CPU. Table 8 gives the run time of every timed final and comparison run: 94 runs and 44.7

Table 8: Run times of the final and comparison runs, from their manifests. Wall-clock minutes per run (data loading, training, forecasting and scoring) and hours in total, on one Google Colab machine with an NVIDIA Tesla T4 (LightGBM uses its CPU). Dominick's: six categories together; a Dominick's world-model run averages three fits. Two M5 runs resumed after the Colab runtime dropped record only the resumed part (the shortest values in their rows). Not counted: runs saved before run times were recorded, the development study, and TimesFM on Dominick's and Frat.

Data	Model	Runs	Median	Range	Hours
M5	World model	9	101.5	37.3 to 103.2	14.0
M5	Lift model	9	88.7	10.3 to 94.6	12.2
M5	LightGBM	9	78.1	76.5 to 79.9	11.7
M5	Graph model	12	7.2	7.2 to 7.5	1.5
M5	TimesFM	8	3.9	3.8 to 4.1	0.5
M5	Seasonal naive	2	1.3	1.2 to 1.3	0.0
Dominick's	World model	18	9.2	9.0 to 10.6	2.8
Dominick's	LightGBM	18	4.6	3.1 to 11.5	1.7
Frat	World model	3	2.2	2.2 to 2.2	0.1
Frat	Lift model	3	2.0	1.9 to 2.0	0.1
Frat	LightGBM	3	1.5	1.5 to 1.6	0.1
All 94 timed runs					44.7

machine-hours in all. A world-model seed on M5 takes about 100 minutes, of which about 3 seconds is forecasting the 28 days; LightGBM about 78 minutes; the graph model about 7 minutes; TimesFM about 4 minutes, with no training. A Dominick's category takes about 9 minutes for the world model (three fits) and 3 to 12 for LightGBM, and a Frat run about 2 minutes. The development study ran on the same machine type and is not counted in the table.

12 Data, code and licence

Data. M5 is available from the competition's Kaggle page [1]; Dominick's from the Kilts Center for Marketing, University of Chicago Booth School of Business [19]; Breakfast at the Frat from dunnhumby [20]. Each is obtained from its provider under the provider's own terms. No data is redistributed with the code; the repository holds loaders that read the providers' files and record their checksums.

Code. https://github.com/Kar488/retail-world-models.

Licence. The work is released for non-commercial use only. The code, including the scripts that draw the figures and tables, is licensed under the PolyForm Noncommercial License 1.0.0. The manuscript text, figures and tables are licensed under Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC 4.0). Commercial use of either needs a separate licence from the author.

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A Supplement: the development study

This section reports what was tried before the final tests, on the development windows only. It shows how the final models were chosen and which ideas from the literature did not help on these data. Every row is a section of the results record, with its run folders.

A.1 Breakfast at the Frat

Benchmarks. On the three windows before the data set's end, seasonal naive scores 0.832, the average of the last 8 weeks 0.756, LightGBM with price only 0.554 and LightGBM with every recorded lever 0.447. The levers are worth a fifth of the error: display, feature and price reduction carry information price does not.

What helped and what did not. Figure 11 shows every configuration screened on Frat. Our item encoder with a direct readout scored 0.424, level with LightGBM (0.421). Building the forecast as a baseline under the nothing-planned plan times a lift (the lift model) brought it to 0.406, better than LightGBM in all five runs and with half the spread across seeds. Letting the 55 items in a store affect each other, by conditional similarity, attention or both, made the forecast worse in every version (0.443 to 0.476) and less steady; limiting it to items in the same category narrowed the gap without closing it. Extra inputs did not help: the plan of the four weeks before (0.411), the same week last year (0.408), the US seasonal calendar (0.425), the count of items on promotion in the store (0.406) and the recorded base price used as the regular price (0.434). The recorded base price changes about twelve times a year per item and equals the shelf price in every week without a promotion, so it is not a slow-moving regular price.

Latent (JEPA-style) losses. Following the JEPA literature [45, 46] and M-JEPA [57], we added a loss that predicts the encoder's own state for the forecast weeks from the state now and the plan, with a slowly updated copy of the encoder as the target [48]. No recipe beat the lift model: alongside the forecast loss 0.410 to 0.411, pretrained on the latent loss and then fine-tuned 0.407 to 0.414. With the encoder frozen after latent-only pretraining the model still reached 0.428, so a state learned without any sales target carries most of the forecast signal. Long pretraining followed by a slowed encoder gave the best display-week error of any configuration (0.813 for the world model, against 0.817 for LightGBM).

The world model. Rolling the state forward week by week under the plan scored 0.408, level with the lift model on the overall score and on every lever. Three ideas from the world-model literature, each at one untuned setting, did not help: keeping the plan's effect apart from the no-plan state as in DWM [53] (0.417), a latent check on each rolled step with a VICReg penalty [49] and near weeks weighted more as in TD-MPC2 [52] (0.427), and all three together (0.424). Training changes did not help either: stopping at the best point on the last 8 training weeks (0.425) took 8 of only 90 training weeks out of the targets, and a term on summed totals in the loss (0.414) had no bias to fix, since total forecasts were within 3% of sales in every run. Readout dropout with weight decay (0.409) was level and steadier, and was kept for the lift model.

Beyond accuracy. Table 9 reports four further tests. *New items*: with one product in ten held out of training, the lift model borrowing the state and level of the ten known items in the same store with the most labels in common scored 0.890 on the new items, against 1.310 for LightGBM fitted on labels, price and plan, and 1.510 for a forecast of zero. On Dominick's analgesics, where items carry no category or maker labels, borrowing failed (2.999 against 0.923): the method needs item attributes. *Combination*: weighting the lift model and LightGBM on a validation window gave the best Frat result of the study (0.388), better than either part on display and circular weeks, because the two make different errors in promoted weeks; on analgesics the combination was level. Combinations were not carried to the final tests, which compare single models. *Plan A against plan B*: for pairs of weeks of the same item and store with different plans, every model picked the week that sold more about 78% of the time; LightGBM was marginally ahead. *Unusual plans*: on lever mixes an item had run in under 2% of its weeks, no model degraded much and LightGBM stayed ahead. The world model's ability to roll a plan forward did not show as an advantage on these tests.

A.2 Dominick's

Analgesics, the category used to set the models. On analgesics, seasonal naive scored 1.274, the average of the last 8 weeks 0.723 and LightGBM 0.715. The lift model scored 0.676 on three seeds, and the world model 0.667 once a totals term was removed from its loss: that term made the total worse, pulling forecasts 9% low (0.819 for the lift model). Training four times longer without a stop broke the summed levels (1.473) while item-by-store error barely moved, so a small bias in the same direction on every item added up at the category total. Five seeds changed the picture: the world model held (0.666) and LightGBM scored 0.702, but the lift model fell to 0.718, so a three-seed figure had flattered it.

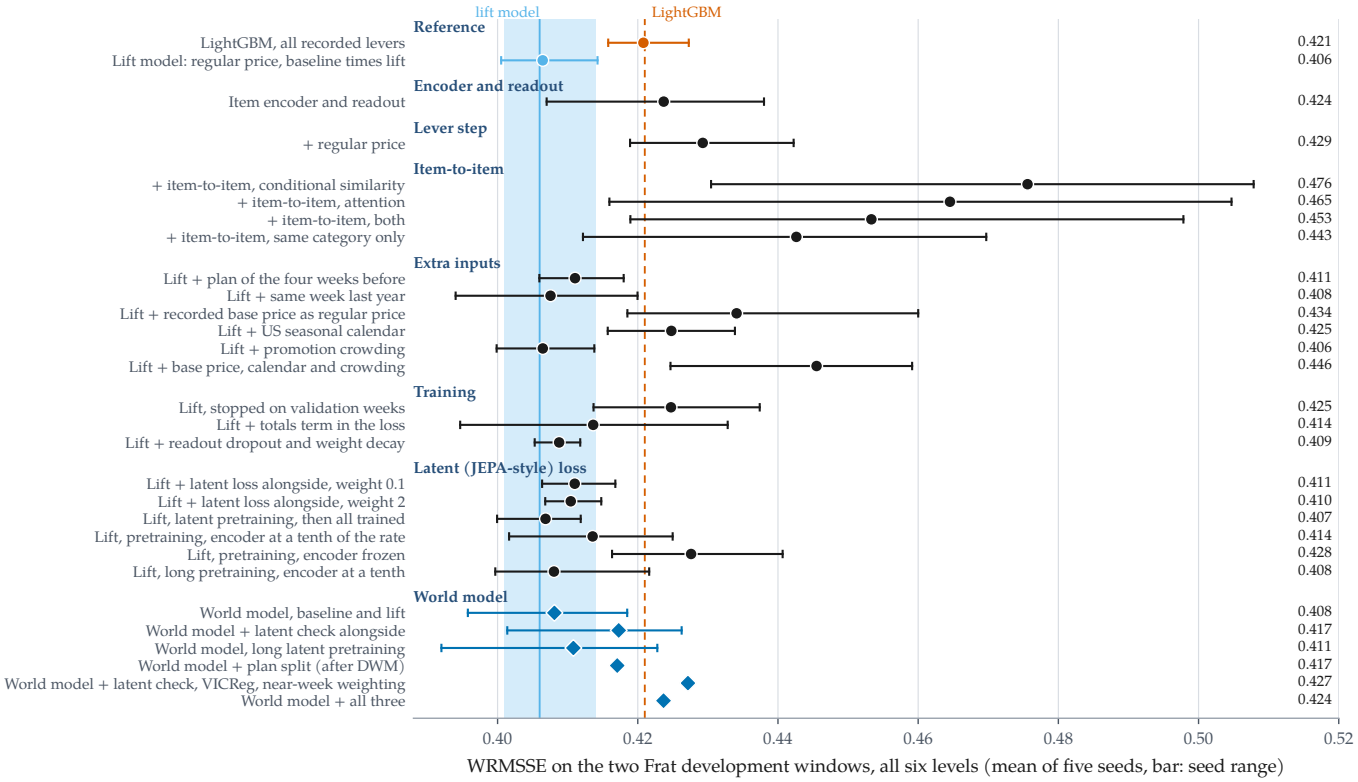


Figure 11: On Frat the gain came from building the forecast as a baseline times a plan lift; item-to-item links, extra inputs, latent losses and the roll-forward did not improve on it. Each row is one configuration on the two development windows, five seeds: mean (marker) and range (bar). Blue band: the lift model's seed range; vertical lines: the lift model (0.406) and LightGBM (0.421). Values below the lift model are in bold. Rows without a bar were recorded as a mean only.

All 28 categories. Run on all 28 categories with settings from analgesics, LightGBM was the most accurate of the three in 14, the lift model in 5 and the world model in 9, with mean WRMSSE 0.688, 0.784 and 0.747; on promoted weeks LightGBM was most accurate in 24 of 27 (Figure 12, Table 10). Analgesics did not generalise, which is why the paper reports every category. The failures (snack crackers 1.463, bath soap 1.465, the lift model on soft drinks 1.406) sat at the category total: item-by-store error was close to LightGBM's. On a six-store sample of snack crackers the cause was in-and-out and seasonal lines: items recorded in 8 to 20 of the last 26 weeks were 8% of sales and the lift model forecast them at 9 times their sales, because the model set each item's level from its 26-week average after the item had fallen away. Capping extreme weeks did little; taking the level from the last 8 weeks removed most of it. On all 28 categories the 8-week level brought the world model's mean to 0.717, removed the large failures (snack crackers 1.463 to 0.874, bath soap 1.465 to 1.195) and gave a little back in categories that had been fine, so it beat LightGBM in 10 categories against 14 before. LightGBM, which sees short recent averages directly, does not have the problem.

Peaks and dips. In cheese, both our model and LightGBM under-forecast promoted weeks by about a third in the holiday window, and cheese peaks do not sit on a fixed week relative to the holidays: they follow the chain's promotion timing, which event flags cannot place. Figure 13 splits forecast against sales into promoted weeks, the weeks just after a promotion, and ordinary weeks, for six categories. The world model did not see the dip after a promotion where the dip is large: it forecast 1.33 times sales after a promotion in snack crackers and 1.32 in soft drinks, against 0.92 and 0.90 for LightGBM. Feeding the model's own sales forecast into the roll-forward, so that a promotion's sales could shape the weeks after it, fixed the soft-drink dip (1.32 to 1.00) and made the overall score worse in four of six categories. Weighting the loss by dollar sales and by how far a week sits from the item's usual level fixed cheese promoted weeks (1.21 to 1.00) and was worse overall in three categories. In frozen entrees every model, LightGBM included, forecast about 0.7 of sales after a promotion: there the week after is stronger than any model expected.

From the first world model to the one tested. Figure 14 follows the world model on the six categories through the changes kept, three seeds each. Across seeds the first

Table 9: What the plan-aware models were also tested on during development: new items, combination with LightGBM, choosing between two plans, and plans an item had rarely run. Development windows, three seeds (Frat combination: five). WRMSSE, lower is better, unless stated. Plan A against plan B: share of week pairs where the forecast picks the week that sold more, and the average miss on the size of the change (log scale).

Data	Model	Result
New items: one product in ten never seen in training (item by store)		
Frat	LightGBM, new item from labels, price and plan	1.310
Frat	Lift model, borrowing from similar items	**0.890**
Frat	Forecast of zero	1.510
Analgesics	LightGBM, from labels, price and plan	**0.923**
Analgesics	Lift model, borrowing (no attribute labels)	2.999
Combination with LightGBM, weights set on a validation window (all levels)		
Frat	Lift model + LightGBM	**0.388**
Frat	World model + LightGBM	0.390
Frat	LightGBM; lift model; world model	0.421; 0.406; 0.408
Analgesics	Lift + LightGBM; world + LightGBM	0.670; 0.668
Plan A against plan B: same item and store, two weeks with different plans (Frat)		
Frat	LightGBM	78.1% right; miss 0.448
Frat	Lift model	77.4% right; miss 0.460
Frat	World model	77.5% right; miss 0.459
Frat	World model + plan split	77.5% right; miss 0.458
Unusual plans: a lever mix the item had in under 2% of its weeks (item by store)		
Frat	LightGBM	0.696 unusual; 0.697 usual
Frat	Lift model	0.742 unusual; 0.725 usual
Frat	World model	0.730 unusual; 0.730 usual

version was unsteady: its range within a category was 0.04 to 0.38, against under 0.03 for LightGBM, and on the mean LightGBM was ahead in five of six. Brand switching fixed what it was built for: in snack crackers and soft drinks, where promoted weeks broke the total, it was better on every seed (0.813 to 0.752, 0.902 to 0.789) and item-by-store error improved too. The fitted share of the summed lift that the category total keeps (Figure 14g) is 0.00 to 0.66 in snack crackers, against about 0.9 in bottled juice, analgesics and frozen entrees: in snack crackers the summed item lifts overstated the category total, and the fit took most of it back. Where it comes out at 1.0 the step changes nothing, a built-in check. The direction agrees with van Heerde, Gupta and Wittink's household estimates [15], but the share is fitted to make totals right; it is not an estimate of switching. Normalising the rolled state narrowed the seed range in five of six categories, and averaging three fits per run with normalisation gave the steadiest version (range 0.05 or less in five categories) and the candidate for the final test: better than LightGBM on the three-seed mean in analgesics (0.621 against 0.701), frozen entrees (0.391 against 0.419) and soft drinks (0.743 against 0.759). One snack-cracker run of the three failed (1.806): its second window's category total broke with promoted weeks forecast at 2.6 times sales while item-by-store error stayed normal, so the summed promotion lift can still run away on one fit in three. Twice the training steps, screened after the final tests were fixed, was better in two categories, worse in two, level in one and mixed in one, and was not kept.

Other changes. Showing the model both its 26-week and its recent levels as inputs, instead of setting its scale by the recent level, was the best world-model version in four of ten categories but did not fix snack crackers (1.515) or soft drinks (1.069): the model did not learn to switch to the recent level when the long level still set its scale. Item role labels (new line, in-and-out, volume driver, long tail, high-low, promotion-responsive, core), assigned from each item's own sales before the windows, moved LightGBM by at most 0.013, since it reads the same information from its history, and moved the world model in both directions; letting the role pick the level broke snack crackers (2.422). We did not call the volume drivers key value items, since a key-value-item list rests on shopper price perception, which these data do not record.

A.3 M5

On the M5 practice window (Table 11), the item encoder with a direct readout scored 0.699 over five seeds against

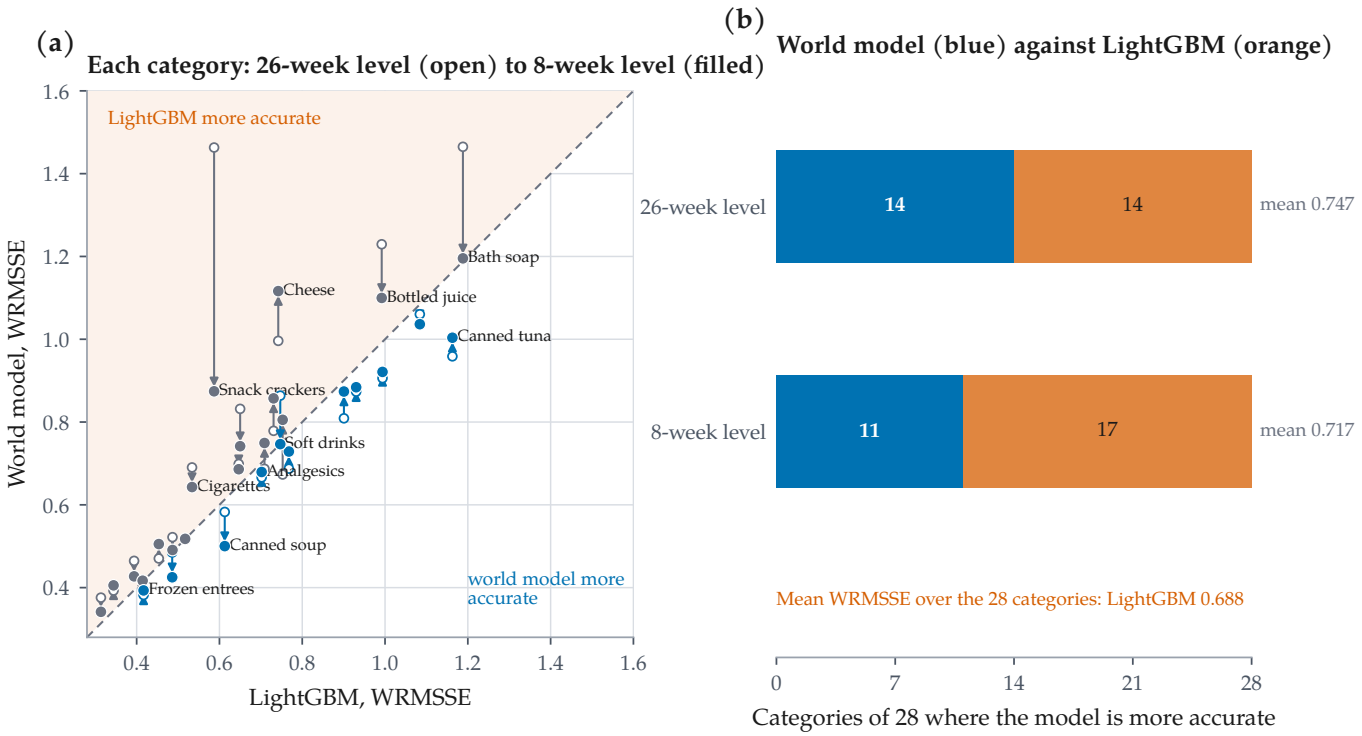


Figure 12: Run unchanged on all 28 Dominick's categories, the world model set on analgesics failed badly in a few; taking each item's level from its last 8 weeks removed the worst failures but did not make it better than LightGBM on average. Development windows, one seed. **(a)** Each category's WRMSSE for LightGBM (horizontal) and the world model (vertical), with the item level from the last 26 weeks (open) and the last 8 weeks (filled); blue: the world model is more accurate with the 8-week level. **(b)** Categories where each model is more accurate, and the mean over the 28.

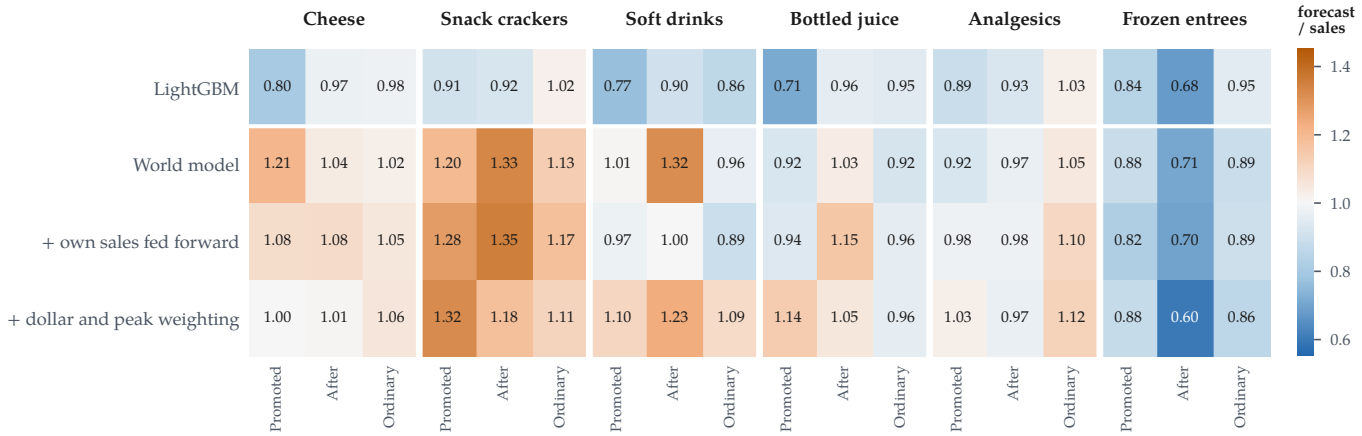


Figure 13: The world model over-forecast the weeks after a promotion where shoppers had stocked up; the two fixes each repaired one category and hurt others. Forecast against sales (1.0 is right; orange over, blue under) in promoted weeks, the weeks just after a promotion and ordinary weeks, Dominick's development windows, one seed. Rows: LightGBM, the world model with the 8-week level, with its own sales forecast fed forward, and with the loss weighted by dollar sales and by peaks.

0.715 for LightGBM, and the lift and world models were level with LightGBM over three seeds (0.713 and 0.707). Normalising the rolled state was the one change that separated the world model: 0.656, with its worst seed (0.675) better than the best seed without it (0.679), all of the gain at the totals and the seed range cut from 0.071 to 0.029. It helped on M5's 28 daily steps and not on Dominick's 8 weekly steps, which fits the reading that it stops the state drifting over a long roll. Averaging three fits did not help on M5 (0.665), where single fits were already steady.

After the official test month, an error analysis of the world model found its 2.6% excess in Foods (+3.9%), in one store (CA_3, +7.5%), and on Sundays and Mondays (+5.1%, +6.4%), not growing with the horizon. Grouped by sales in the 28 days before the test, the ex-

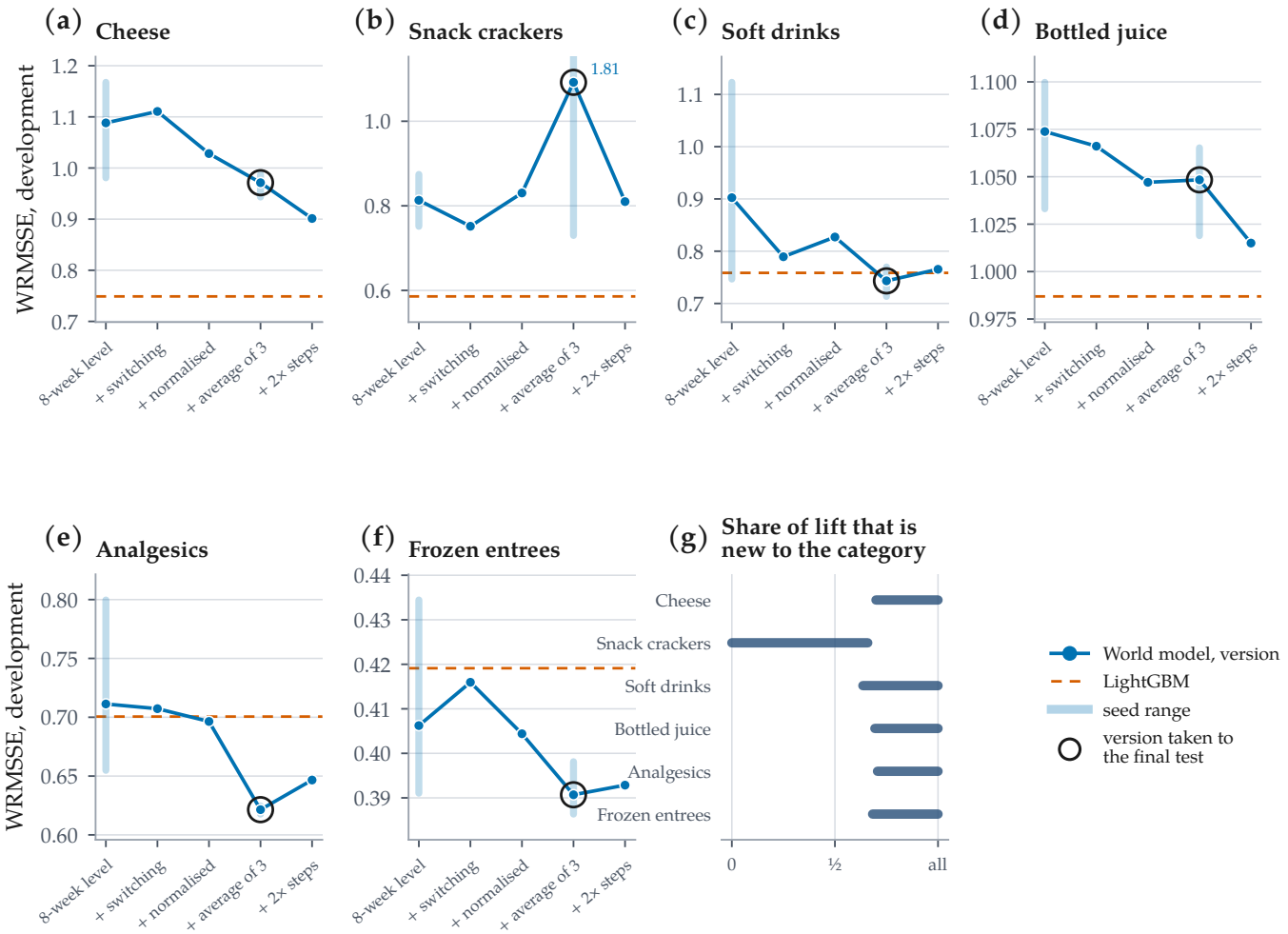


Figure 14: Brand switching, a normalised roll-forward and averaging three fits took the world model past LightGBM in three of six categories on the development windows, and the switching step learned how much of a promotion's lift is new to the category. **(a–f)** Development WRMSSE of each version, three-seed means; shaded bars: seed range of the first version and of the one taken to the final test (circled); dashed: LightGBM. **(g)** The fitted share of the summed promotion lift that the category total keeps, the split that makes category totals fit best (not an estimate of switching), range over runs.

cess came from items that had been selling fast (+5.4% and +6.1% on items selling 3 to 10 and over 10 a day). Four changes were screened on the practice window under rules written before screening: weekday as seven inputs, event kinds, twice the training steps, and the item's level against its past-year level. None beat the original (Table 11), so there was no second version. On the practice window every version under-forecast by 4% to 6%, while on the test month the world model over-forecast by 2.6%: the sign of the level error changed between adjacent months, so a fix tuned on one would not carry to the next. That finding is what led to the confirmation months of §6.2.

Table 10: Across all 28 Dominick's categories LightGBM is more accurate on average; the world model wins in 14 with its first level and 10 with the 8-week level. Development windows, one seed (analgesics: five), settings set on analgesics and not tuned per category. World model with the item's level from its last 26 weeks or last 8 weeks. Lowest in each row in bold.

Category	LightGBM	Lift	World, 26 wk	World, 8 wk
Analgesics	0.702	0.718	**0.666**	0.679
Bath soap	**1.188**	1.568	1.465	1.195
Beer	**0.454**	0.460	0.470	0.505
Bottled juice	**0.992**	1.247	1.229	1.100
Canned soup	0.613	0.532	0.583	**0.500**
Canned tuna	1.163	1.030	**0.959**	1.004
Cereals	0.753	0.795	**0.673**	0.805
Cheese	**0.742**	0.900	0.996	1.116
Cigarettes	**0.534**	0.918	0.690	0.643
Cookies	**0.487**	0.503	0.522	0.491
Crackers	0.709	0.967	**0.687**	0.749
Dish detergent	**0.344**	0.436	0.393	0.405
Fabric softener	0.414	**0.358**	0.397	0.417
Front-end candies	**0.650**	0.748	0.832	0.742
Frozen dinners	0.930	0.905	**0.873**	0.884
Frozen entrees	0.417	**0.374**	0.382	0.393
Frozen juices	**0.314**	0.360	0.376	0.342
Grooming products	0.901	1.208	**0.809**	0.874
Laundry detergent	0.768	0.735	**0.687**	0.729
Oatmeal	1.084	**1.026**	1.060	1.036
Paper towels	0.486	0.445	0.485	**0.425**
Shampoos	0.994	0.938	**0.906**	0.921
Snack crackers	**0.587**	0.959	1.463	0.874
Soaps	0.517	0.521	**0.516**	0.518
Soft drinks	0.747	1.406	0.864	**0.747**
Toilet tissue	**0.394**	0.413	0.465	0.427
Toothbrushes	**0.647**	0.671	0.699	0.686
Toothpaste	**0.731**	0.802	0.779	0.857
Mean	0.688	0.784	0.747	0.717

Table 11: On the M5 practice window, only the normalised roll-forward separated the world model from LightGBM; no later change beat it. WRMSSE over all 12 levels, 25 April to 22 May 2016. A screened change was kept only if it lowered the score, and a change that doubles the cost only if by more than the seed range (0.029).

Model or change	Mean	Seed range
Seasonal naive	0.870	
LightGBM, one model per store	0.715	
Lift model (3 seeds)	0.713	0.683 to 0.734
Item encoder and readout (5 seeds)	0.699	0.669 to 0.724
World model (3 seeds)	0.707	0.679 to 0.750
World model + normalised roll-forward (3 seeds)	0.656	0.646 to 0.675
+ average of three fits (1 run)	0.665	
+ weekday as seven inputs (seed 1)	0.653	
+ weekday and event kinds (seed 1)	0.646	
+ 60,000 training steps (seed 1)	0.642	
+ item level against past-year level (seed 1)	0.680	